

Excellent. These next questions are the ones **senior AWS Data Engineer interviewers (5-8 years)** ask. They are mostly **real production scenarios** rather than definitions. Answer them like you're explaining what happened in your project.

AWS Data Engineer Interview (Q51-Q100)

Production Scenario Based

Q51. A Glue Job suddenly failed in Production. What would you do?

Interview Answer (2-3 mins)

"The first thing I do is avoid rerunning the job immediately. I first identify the root cause.

My approach is:

1. Check CloudWatch logs for the Glue Job.
2. Identify whether it is a source issue, permission issue, schema issue, or resource issue.
3. Verify whether the source file arrived completely in S3.
4. Check whether the Glue Crawler updated the schema.
5. Validate IAM permissions.
6. Verify Spark executor errors.
7. Check whether the target S3 bucket has enough permissions.
8. After fixing the issue, rerun only the failed partition instead of the complete job.

In our investment banking project, a Glue job failed because a source system added two new columns unexpectedly. The Glue mapping did not recognize the new schema. We updated the mapping, reran the failed partition, and validated the downstream Athena reports."

Q52. Athena query suddenly became slow. What did you check?

Answer

"I checked the following:

- Whether partition pruning was working
- Whether users queried the partition column
- File format (CSV or Parquet)
- Number of small files
- Whether the Glue Catalog had outdated metadata
- Query execution statistics
- Data scanned

In one case, users forgot to filter by TradeDate. Athena scanned nearly 4 TB instead of one day's partition, causing slow performance and higher cost. After adding the partition filter, execution time dropped significantly."

Q53. Lambda function timed out. How did you solve it?

Answer

"Lambda has a maximum execution timeout. I first checked CloudWatch Logs to determine where execution stopped.

I verified:

- Function duration
- Memory usage
- Network calls
- S3 read/write latency
- Payload size

If the function was doing heavy processing, I moved that logic into AWS Glue and kept Lambda responsible only for validation and workflow triggering.

This reduced execution time and avoided timeout errors."

Q54. Users reported missing records in Athena. What did you investigate?

Answer

"I checked:

- Whether the files were uploaded to S3
- Glue Job completion
- Glue Crawler updates
- Partition registration
- Athena table location
- Data quality logs

In one case, the data existed in S3 but the crawler had not registered the new partition. After repairing the table and updating partitions, the records became visible."

Q55. S3 Access Denied Error. What could be the reasons?

Answer

"I checked:

- IAM Role permissions
- Bucket Policy
- KMS encryption permissions
- Object ACLs
- Cross-account permissions
- VPC Endpoint policies

Most S3 Access Denied issues are caused by missing IAM permissions or restrictive bucket policies."

Q56. How do you secure your S3 bucket?

Answer

"We followed several best practices:

- Block Public Access
- IAM least-privilege access
- Bucket policies
- Server-side encryption (SSE-KMS)
- Versioning
- MFA Delete for sensitive buckets
- CloudTrail auditing
- Lifecycle policies

- Access logging"
-

Q57. How do you optimize Glue Job performance?

Answer

"We optimized Glue jobs by:

- Reading only required columns
 - Applying filters early
 - Converting CSV to Parquet
 - Partitioning data
 - Repartitioning before writing
 - Using incremental loads
 - Increasing DPU's only when necessary
 - Avoiding unnecessary joins and shuffles"
-

Q58. How do you optimize S3 costs?

Answer

"We optimized costs by:

- Using lifecycle policies
 - Compressing files
 - Converting CSV to Parquet
 - Deleting temporary files
 - Moving old data to Glacier
 - Avoiding duplicate storage
 - Using partition pruning"
-

Q59. Explain your S3 folder structure.

Answer

```
s3://bank-project/
```

```
raw/
```

processed/

curated/

archive/

error/

logs/

backup/

Raw stores source files.

Processed stores cleaned data.

Curated stores business-ready Parquet files.

Archive stores historical data.

Error stores rejected files.

Logs store Glue and validation reports.

Q60. How do you manage failed files?

Answer

"When validation fails, we don't delete the file.

Instead:

Move to Error Folder

↓

Generate Validation Report

↓

Send SNS Email

↓

Notify Source Team

↓

Correct File



Reload"

Q61. Explain watermarking.

Answer

"A watermark stores the maximum processed timestamp.

Example:

Yesterday processed until

2026-07-28 22:15

Today's query loads only data after that timestamp.

This avoids duplicate processing."

Q62. How do you avoid duplicate loads?

Answer

"We used:

- TradeID
 - File checksum
 - LastUpdatedTimestamp
 - S3 filename validation
 - Spark dropDuplicates()
 - MERGE logic where applicable"
-

Q63. Explain CDC (Change Data Capture).

Answer

"CDC captures only inserts, updates, and deletes instead of reading the full table.

Benefits:

- Faster processing

- Lower cost
- Less network traffic
- Near real-time updates

CDC is commonly implemented using database logs or timestamps."

Q64. Difference between CDC and Incremental Load?

Answer

Incremental:

- New records only (or based on watermark)

CDC:

- Inserts
- Updates
- Deletes

CDC captures all changes, while incremental loads often focus on new or modified data depending on the implementation.

Q65. Explain Slowly Changing Dimensions (SCD).

Answer

"SCD manages historical changes in dimension tables.

Example:

Customer changes address.

Business may need:

- Overwrite the old value (Type 1)
 - Keep history (Type 2)
 - Preserve selected historical values (Type 3)"
-

Q66. Difference between SCD Type 1 and Type 2?

Answer

Type 1:

- Overwrites existing value
- No history

Type 2:

- Creates a new record
- Maintains history
- Uses effective dates and current flags

For banking customer data, we usually prefer Type 2 to retain audit history.

Q67. How do you handle bad records?

Answer

"We never stop the entire pipeline because of a few bad records.

Instead:

- Validate schema
 - Separate invalid rows
 - Store them in an error bucket
 - Continue processing valid records
 - Generate an error report for investigation"
-

Q68. What is Data Quality Validation?

Answer

"We validate:

- Null values
- Duplicate keys
- Invalid dates
- Mandatory columns
- Data types
- Business rules

- Referential integrity
 - File counts"
-

Q69. How do you reconcile source and target data?

Answer

"We compare:

- Record counts
- Aggregate totals
- Duplicate counts
- Null counts
- Hash totals
- Sample records

Reconciliation ensures no data loss during ETL."

Q70. Explain Hash Validation.

Answer

"We generate hash values using business columns.

If source and target hashes match, the data is considered unchanged.

Hashing is useful for detecting updates efficiently."

Q71. Why partition by TradeDate?

Answer

"Most business queries filter by date.

Partitioning by TradeDate minimizes data scanned, improves Athena and Spark performance, and reduces query cost."

Q72. Why not partition by CustomerID?

Answer

"CustomerID has very high cardinality, creating millions of tiny partitions. This increases metadata overhead and hurts query performance.

Partition columns should have manageable cardinality and match common query filters."

Q73. Explain the Small File Problem.

Answer

"Thousands of small files increase metadata, create many Spark tasks, and slow Athena.

We periodically compacted files to around 128–256 MB using repartitioning or compaction jobs."

Q74. What if one partition contains 90% of the data?

Answer

"That indicates data skew.

Solutions include:

- Salting keys
 - Better partition keys
 - Repartitioning
 - Adaptive Query Execution
 - Balancing Spark partitions"
-

Q75. What is Data Skew?

Answer

"Data skew occurs when one partition contains significantly more data than others.

Effects:

- Long-running tasks
- Executor imbalance
- Poor cluster utilisation

We identify skew through Spark UI and rebalance the data."

Q76. Explain Spark Shuffle.

Answer

"Shuffle redistributes data across executors during joins, aggregations, or repartitioning.

It is expensive because it involves disk I/O and network transfer.

We minimise unnecessary shuffles by filtering early and choosing appropriate partitioning."

Q77. Why convert CSV to Parquet?

Answer

"Parquet reduces storage, supports compression, enables column pruning, and improves Athena, Glue, and EMR query performance."

Q78. How do you monitor AWS resources?

Answer

"We use CloudWatch for:

- Metrics
- Logs
- Dashboards
- Alarms

CloudTrail records API activity for auditing and security."

Q79. Explain CloudTrail.

Answer

"CloudTrail records AWS API calls such as who created, modified, or deleted resources.

It is primarily used for auditing, compliance, and security investigations."

Q80. Difference between CloudWatch and CloudTrail?

Answer

CloudWatch:

- Metrics
- Logs
- Monitoring
- Alarms

CloudTrail:

- API history
 - User activity
 - Audit trail
 - Compliance
-

Q81. Explain Glue Bookmark.

Answer

"Glue Bookmarks track previously processed data, allowing subsequent runs to process only new files or records. This is especially useful for incremental ETL."

Q82. What happens if a Glue Bookmark is reset?

Answer

"If a bookmark is reset, the job treats all eligible input as new and may reprocess previously loaded data unless additional deduplication logic exists."

Q83. How do you handle late-arriving files?

Answer

"We identify late files using business dates and arrival timestamps, process them separately if needed, update the affected partition, and rerun downstream validation."

Q84. How do you manage schema evolution?

Answer

"We allow controlled schema evolution through Glue Crawlers, version ETL mappings, validate downstream compatibility, and test changes before production deployment."

Q85. What is idempotency in ETL?

Answer

"An idempotent ETL job can be run multiple times with the same input without creating duplicate or inconsistent data. We achieve this using unique business keys, merge logic, and deduplication."

Q86. How do you tune Redshift queries?

Answer

"We optimise distribution keys, sort keys, compression encoding, VACUUM, ANALYZE, and avoid unnecessary data movement across nodes."

Q87. What is VACUUM in Redshift?

Answer

"VACUUM reorganises table storage after updates and deletes, improving query performance and reclaiming space."

Q88. What is ANALYZE in Redshift?

Answer

"ANALYZE updates table statistics so the query optimiser can generate efficient execution plans."

Q89. How do you secure data in AWS?

Answer

"We use IAM roles, KMS encryption, VPC endpoints, private subnets, CloudTrail, CloudWatch alarms, bucket policies, and least-privilege access."

Q90. Explain KMS.

Answer

"AWS Key Management Service (KMS) manages encryption keys used to encrypt S3 objects, Redshift data, EBS volumes, and other AWS resources securely."

Q91. Describe a major production incident you resolved.

Answer

"A schema change from an upstream trading application caused the Glue ETL job to fail. We identified the issue from CloudWatch logs, validated the file against the expected schema, coordinated with the source team, updated the ETL mapping, reran the failed partition, reconciled the output using Athena, and confirmed downstream reports were accurate."

Q92. How do you minimise AWS costs?

Answer

"We use incremental processing, Parquet with compression, partition pruning, lifecycle policies, right-sized Glue workers, auto-scaling where appropriate, and

archival to Glacier."

Q93. What KPIs do you monitor?

Answer

"We monitor job success rate, processing time, failed file count, data latency, data quality issues, Athena data scanned, Redshift load duration, and CloudWatch alarms."

Q94. How do you ensure data consistency?

Answer

"We implement reconciliation checks, transactional loading where possible, retry mechanisms, idempotent processing, and post-load validation before publishing data."

Q95. What happens if an S3 file is uploaded twice?

Answer

"We validate file names, checksums, and business keys before processing. Duplicate files are quarantined or skipped, preventing duplicate downstream records."

Q96. How do you restart a failed pipeline?

Answer

"We restart from the failed stage rather than rerunning the entire workflow. We use bookmarks, watermarks, and partition-level reruns to minimise processing time."

Q97. What documentation do you maintain?

Answer

"We maintain architecture diagrams, data flow documents, source-to-target mappings, runbooks, operational procedures, IAM documentation, recovery procedures, and data dictionaries."

Q98. What are the biggest challenges in investment banking ETL?

Answer

"The main challenges are strict data accuracy, low-latency processing, audit requirements, schema changes, large transaction volumes, reconciliation, and regulatory compliance."

Q99. If you become the lead engineer, what improvements would you introduce?

Answer

"I would strengthen automated data quality checks, implement CI/CD for ETL deployments, improve monitoring dashboards, automate schema validation, optimise costs continuously, and standardise operational runbooks."

Q100. Why should we hire you?

Answer

"I have hands-on experience designing and supporting AWS-based data pipelines using S3, Glue, Lambda, Athena, Redshift, EMR, IAM, CloudWatch, and the Glue Data Catalog. Beyond building pipelines, I understand production support, troubleshooting, performance tuning, data quality, security, and cost optimisation. I can quickly identify root causes during production incidents, collaborate with upstream and downstream teams, and deliver reliable, scalable data solutions that meet business and regulatory requirements. These are the skills I would bring to your data engineering team."

These 100 questions form a solid interview preparation set. An excellent follow-up would be **Q101-Q200**, focusing exclusively on advanced AWS Glue, Spark,

PySpark, IAM, S3, Athena, Redshift, EMR, Snowflake, Airflow, SQL, and scenario-based questions commonly asked for senior AWS Data Engineer roles.

Excellent. These are **senior-level AWS Data Engineer interview questions (Q161-180)** focused on **Amazon S3, IAM, Athena, Redshift, EMR, Lake Formation, Iceberg, Delta Lake, and production architecture**. These are commonly asked in **Goldman Sachs, JP Morgan, Morgan Stanley, Barclays, UBS, Deloitte, Cognizant, Accenture, Capgemini**, and similar interviews.

AWS Data Engineer Interview Part-6 (Q161-180)

Q161. Explain Amazon S3 Architecture in your project.

Interview Answer (3 Minutes)

"In our investment banking project, Amazon S3 acted as the enterprise data lake. We organised data into logical layers:

- **Raw Layer** – Original CSV and JSON files received from upstream systems.
- **Processed Layer** – Data after validation and cleansing.
- **Curated Layer** – Business-ready Parquet files partitioned by TradeDate.
- **Archive Layer** – Historical data retained for compliance.
- **Error Layer** – Invalid files with validation reports.

AWS Lambda validated incoming files, AWS Glue transformed them, Athena queried the curated layer, and Amazon Redshift consumed aggregated datasets for reporting.

We enabled versioning, server-side encryption with KMS, lifecycle policies, and IAM-based access control to secure and manage the data."

Q162. Explain S3 Versioning.

Interview Answer

"S3 Versioning preserves multiple versions of the same object.

If a file is overwritten or accidentally deleted, the previous version can be restored.

In financial systems, versioning helps with:

- Recovery from accidental deletion
- Audit requirements
- Regulatory compliance
- Rollback after incorrect uploads

We enabled versioning for critical production buckets."

Q163. Explain S3 Lifecycle Policy.

Interview Answer

"S3 Lifecycle Policies automatically transition objects between storage classes or delete them after a defined period.

Example:

- First 30 days → S3 Standard
- After 30 days → Standard-IA
- After 90 days → Glacier Flexible Retrieval
- After 365 days → Glacier Deep Archive

This reduced storage costs while meeting compliance requirements."

Q164. What is Multipart Upload?

Interview Answer

"Multipart Upload divides a large file into smaller parts that are uploaded in parallel.

Benefits:

- Faster uploads
- Automatic retry of failed parts
- Better network utilisation
- Support for very large files

We used multipart uploads for large daily trade files."

Q165. Explain Cross-Region Replication (CRR).

Interview Answer

"Cross-Region Replication automatically copies objects from one S3 bucket to another bucket in a different AWS Region.

We use CRR for:

- Disaster recovery
- Business continuity
- Compliance
- Low-latency access in another Region

Replication requires versioning to be enabled on both buckets."

Q166. Explain Cross-Account IAM Role.

Interview Answer

"In large organisations, different AWS accounts are used for development, testing, production, and data sharing.

Instead of creating IAM users in every account, we configure an IAM Role in the target account with a trust policy. Users or services in another account assume that role using AWS STS temporary credentials.

This is more secure than sharing long-lived access keys."

Q167. What is AWS STS?

Interview Answer

"AWS Security Token Service (STS) provides temporary security credentials.

Advantages:

- No long-term credentials
- Temporary access
- Supports AssumeRole

- Improved security
 - Commonly used for cross-account access"
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Q168. Explain AssumeRole.

Interview Answer

"AssumeRole allows a user or AWS service to temporarily use the permissions of another IAM role.

Example:

A Glue job running in Account A assumes a role in Account B to read data from an S3 bucket.

This supports secure cross-account data access without sharing credentials."

Q169. Explain Permission Boundaries.

Interview Answer

"Permission Boundaries define the maximum permissions an IAM user or role can receive.

Even if an IAM policy grants broad access, the boundary prevents permissions beyond the configured limit.

This helps organisations enforce governance and prevent privilege escalation."

Q170. Explain Athena CTAS (Create Table As Select).

Interview Answer

"CTAS creates a new table from the results of a SELECT query.

We used CTAS to:

- Convert CSV into Parquet
- Repartition data
- Aggregate datasets
- Create optimised reporting tables

CTAS is useful because it combines transformation and table creation in a single operation."

Q171. How do you optimise Athena performance?

Interview Answer

"We optimise Athena by:

- Using Parquet or ORC instead of CSV
 - Partitioning by frequently filtered columns
 - Enabling partition pruning
 - Compressing files with Snappy
 - Avoiding SELECT *
 - Compacting small files
 - Reading only required columns
 - Maintaining accurate Glue Catalog metadata"
-

Q172. Explain Partition Projection in Athena.

Interview Answer

"Partition Projection allows Athena to calculate partition values instead of storing every partition in the Glue Catalog.

Benefits:

- Faster partition discovery
 - Reduced metadata management
 - Better scalability for very large datasets
 - Lower operational overhead"
-

Q173. Explain Redshift Distribution Styles.

Interview Answer

"Distribution determines how data is spread across Redshift nodes.

Common styles:

- AUTO – Redshift chooses automatically.
- EVEN – Rows distributed evenly.
- KEY – Rows distributed based on a selected column.
- ALL – Entire table copied to every node.

For large fact tables joined on CustomerID, we often used KEY distribution on CustomerID to minimise data movement."

Q174. Explain Sort Keys.

Interview Answer

"Sort Keys define the physical ordering of rows in Redshift.

If queries frequently filter by TradeDate, using TradeDate as a Sort Key allows Redshift to skip unnecessary blocks during scans.

This improves query performance significantly."

Q175. Explain Materialized Views.

Interview Answer

"A Materialized View stores the result of a query physically on disk.

Unlike a standard view, it does not execute the underlying query each time.

We used Materialized Views for frequently accessed aggregated reports to reduce query execution time."

Q176. Explain Amazon EMR Cluster Types.

Interview Answer

"EMR supports:

- Long-running clusters for continuous processing
- Transient clusters that terminate after jobs complete

We preferred transient clusters for scheduled ETL because they reduced infrastructure costs by shutting down automatically after processing."

Q177. Difference between Glue and EMR.

Interview Answer

AWS Glue	Amazon EMR
Serverless ETL	Managed big data cluster
AWS manages infrastructure	Customer controls cluster
Best for standard ETL	Best for complex Spark/Hadoop workloads
Limited Spark tuning	Full Spark and cluster configuration
Quick setup	Greater flexibility

"For routine ETL we used Glue, while EMR was chosen for large-scale Spark processing that required custom tuning."

Q178. What is AWS Lake Formation?

Interview Answer

"AWS Lake Formation simplifies the creation, security, and governance of data lakes.

It provides:

- Fine-grained table and column permissions
- Centralised access control
- Integration with Glue Data Catalog
- Data governance and auditing

It reduces the need to manage complex IAM and S3 bucket policies for data access."

Q179. Difference between Delta Lake, Apache Iceberg, and Apache Hudi.

Interview Answer

Feature	Delta Lake	Apache Iceberg	Apache Hudi
ACID Transactions	Yes	Yes	Yes
Time Travel	Yes	Yes	Yes
Schema Evolution	Yes	Yes	Yes
Streaming Support	Strong	Good	Strong
Ecosystem	Spark-first	Multi-engine	Spark-focused

"All three bring database-like capabilities to data lakes, including ACID transactions, schema evolution, and time travel. The choice depends on the analytics engines and ecosystem used by the organisation."

Q180. Describe a major production incident involving S3, Glue, and Athena.

Interview Answer (3 Minutes)

"One morning, business users reported that previous day's trade reports were missing data in Athena.

I first checked whether the source files had arrived in the S3 Raw bucket. The files were present. Next, I verified the Lambda logs and confirmed validation had completed successfully.

I then reviewed the Glue Workflow and found that the ETL job had completed, but the Glue Crawler had failed due to an IAM permission issue after a policy update. Because the crawler did not update the Glue Data Catalog, the new partition was not registered, so Athena could not see the latest data.

To resolve the issue, I restored the required IAM permissions, reran the crawler, verified the new partition in the Glue Catalog, and re-executed the Athena validation queries. After confirming the expected record counts matched the source, I informed the reporting team that the data was available.

To prevent recurrence, we added CloudWatch alarms for crawler failures and included IAM policy validation in our deployment checklist."

Next Part (Q181-200): Senior Production & Architecture Scenarios

The final section will cover the types of questions often used in **L3/L4 AWS Data Engineer interviews**, including:

- End-to-end architecture design
- Disaster recovery and backup strategy
- High availability and fault tolerance
- Cost optimisation across AWS services
- Security and compliance (IAM, KMS, Lake Formation)
- Real production incidents and root-cause analysis
- Performance tuning scenarios
- Investment banking ETL troubleshooting
- CI/CD for Glue and infrastructure
- Behavioural and leadership questions with technical depth

181-200

Excellent. **Q181-Q200** are the kinds of questions that distinguish a **Senior AWS Data Engineer (6-10 years)** from a mid-level engineer. These focus on **architecture, disaster recovery, performance, security, production incidents, leadership, and investment banking scenarios**.

AWS Data Engineer Interview Part-7 (Q181-Q200)

Q181. Design an AWS Data Lake Architecture for an Investment Banking Project.

Interview Answer (3-4 minutes)

"I would design a layered architecture to ensure scalability, security, and maintainability.

1. **Source Systems** – Trading platforms, market data feeds, payment systems, and reference data.

- 2. Ingestion Layer** – Files land in Amazon S3 Raw through SFTP, APIs, or streaming services.
 - 3. Validation Layer** – AWS Lambda validates file names, schema, file size, and duplicates.
 - 4. Metadata Layer** – AWS Glue Crawlers update the Glue Data Catalog.
 - 5. Processing Layer** – AWS Glue or Amazon EMR performs cleansing, deduplication, joins, and business transformations.
 - 6. Storage Layer** – Curated data is stored as partitioned Parquet files in S3.
 - 7. Analytics Layer** – Amazon Athena for ad-hoc queries and Amazon Redshift for reporting.
 - 8. Monitoring** – CloudWatch dashboards, SNS alerts, and CloudTrail auditing.
 - 9. Security** – IAM roles, KMS encryption, Lake Formation, and VPC endpoints."
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Q182. How do you make an ETL pipeline fault tolerant?

Interview Answer

"We make pipelines fault tolerant by:

- Using retry mechanisms for transient failures.
 - Designing ETL jobs to be idempotent.
 - Processing data incrementally with bookmarks or watermarks.
 - Writing outputs only after successful completion.
 - Logging detailed execution information.
 - Sending CloudWatch/SNS alerts.
 - Supporting restart from the failed stage instead of restarting the entire workflow."
-

Q183. How do you ensure High Availability (HA)?

Interview Answer

"We use:

- Amazon S3 for highly durable storage.
- Multi-AZ services where applicable.
- Redundant ETL scheduling.
- Cross-Region Replication for critical buckets.

- Multiple availability zones for databases.
 - Auto Scaling where supported.
 - Health monitoring with CloudWatch.
 - Disaster recovery procedures and backups."
-

Q184. What is your Disaster Recovery (DR) strategy?

Interview Answer

"Our DR strategy includes:

- S3 Cross-Region Replication for important data.
 - Redshift snapshots.
 - Infrastructure as Code using Terraform or CloudFormation.
 - Backup of Glue scripts and metadata.
 - Recovery runbooks.
 - Periodic DR testing to verify recovery objectives."
-

Q185. How do you secure sensitive banking data?

Interview Answer

"We apply multiple security layers:

- IAM least-privilege access.
 - AWS KMS encryption at rest.
 - TLS encryption in transit.
 - Secrets Manager for credentials.
 - Lake Formation for fine-grained access.
 - CloudTrail for auditing.
 - CloudWatch for monitoring.
 - Data masking where required.
 - VPC endpoints to avoid public internet access."
-

Q186. How do you handle Personally Identifiable Information (PII)?

Interview Answer

"For PII:

- Encrypt sensitive fields.
 - Restrict access through IAM and Lake Formation.
 - Mask or tokenize data for non-production environments.
 - Log all access through CloudTrail.
 - Follow data retention and compliance policies."
-

Q187. What if the source system sends corrupted files?

Interview Answer

"Our validation layer checks:

- File format.
- Schema.
- Mandatory columns.
- Record count.
- File size.
- Checksums if available.

Invalid files are moved to an error bucket, logged, and the source team is notified. Valid files continue through the pipeline to avoid unnecessary delays."

Q188. How do you reduce AWS costs?

Interview Answer

"We reduce costs by:

- Using incremental loads.
 - Converting CSV to Parquet.
 - Partitioning datasets.
 - Compressing files.
 - Applying lifecycle policies.
 - Choosing appropriate Glue worker types.
 - Using transient EMR clusters.
 - Eliminating unnecessary full refreshes.
 - Monitoring Athena data scanned."
-

Q189. Explain your deployment process.

Interview Answer

"Our deployment process includes:

1. Development and unit testing.
 2. Code review.
 3. CI/CD pipeline execution.
 4. Deployment to Development.
 5. QA/UAT validation.
 6. Production approval.
 7. Production deployment during a maintenance window.
 8. Post-deployment validation.
 9. Rollback plan if issues occur."
-

Q190. How do you monitor production?

Interview Answer

"We monitor:

- Glue job status.
 - Lambda execution.
 - CloudWatch metrics and logs.
 - Athena query performance.
 - Redshift load duration.
 - Failed file count.
 - Data quality metrics.
 - CloudTrail security events.
 - SNS alerts for failures."
-

Q191. A Glue Job finishes successfully, but the business says data is incorrect. What do you do?

Interview Answer

"I don't assume the ETL is correct just because the job succeeded.

I:

1. Compare source and target record counts.
 2. Validate business rules.
 3. Review transformation logic.
 4. Check joins and filters.
 5. Verify partition locations.
 6. Run reconciliation queries in Athena.
 7. Compare sample records with the source.
 8. If a logic issue exists, fix the ETL, rerun the affected partition, and revalidate before releasing the data."
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Q192. Describe a difficult production issue you solved.

Interview Answer

"We once had an overnight trade load where record counts were significantly lower than expected.

Investigation showed the Glue job had completed successfully, but a filter condition incorrectly excluded trades with a newly introduced transaction status.

After reviewing the transformation logic and confirming the business requirement with stakeholders, we updated the ETL, reran the affected partition, reconciled the data, and implemented regression tests to prevent similar issues."

Q193. How do you improve pipeline reliability?

Interview Answer

"I improve reliability by:

- Adding input validation.
- Implementing retries.
- Using idempotent processing.
- Enabling detailed logging.
- Creating CloudWatch alarms.
- Monitoring data quality.
- Automating health checks.

- Maintaining runbooks and operational documentation."
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Q194. How do you optimise large joins?

Interview Answer

"I:

- Broadcast small reference tables.
 - Filter data before joins.
 - Select only required columns.
 - Repartition on join keys where appropriate.
 - Enable Adaptive Query Execution.
 - Monitor shuffle metrics.
 - Address data skew using better partitioning or salting."
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Q195. How do you investigate a slow ETL job?

Interview Answer

"My investigation includes:

- Reviewing Spark UI.
 - Checking executor utilisation.
 - Looking for skewed partitions.
 - Measuring shuffle read/write.
 - Reviewing recent code changes.
 - Checking input data volume.
 - Ensuring partition pruning is working.
 - Validating worker sizing.
 - Comparing execution with previous successful runs."
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Q196. Explain your data quality framework.

Interview Answer

"Our data quality framework validates:

- Null values.
- Duplicate keys.
- Data types.

- Mandatory columns.
 - Referential integrity.
 - Record counts.
 - Business rules.
 - Reconciliation between source and target.
 - Error reporting and exception handling."
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Q197. How do you communicate during a production incident?

Interview Answer

"During an incident:

1. Acknowledge the issue quickly.
 2. Assess business impact.
 3. Communicate regular status updates.
 4. Coordinate with source, infrastructure, and business teams.
 5. Resolve the issue.
 6. Validate the fix.
 7. Perform a root cause analysis.
 8. Document preventive actions."
-

Q198. What would you automate in your project?

Interview Answer

"I would automate:

- File validation.
 - Schema validation.
 - Data quality checks.
 - Glue job execution.
 - Deployment through CI/CD.
 - Infrastructure provisioning with Terraform.
 - Cost monitoring.
 - Operational dashboards.
 - Alerting and reporting."
-

Q199. Why should we choose you as a Senior AWS Data Engineer?

Interview Answer

"I have practical experience designing, building, and supporting AWS data platforms using Amazon S3, AWS Glue, Lambda, Athena, Redshift, EMR, IAM, CloudWatch, and the Glue Data Catalog. Beyond development, I have worked on production support, troubleshooting, performance tuning, security, and cost optimisation. I focus on delivering reliable, scalable ETL pipelines while collaborating effectively with business users, operations teams, and upstream application owners."

Q200. Tell me about yourself. (Senior AWS Data Engineer)

Interview Answer (3-4 minutes)

"Hello, my name is **Ramasamy**.

I have experience as an AWS Data Engineer working on investment banking data platforms. My primary responsibility has been designing and supporting end-to-end ETL pipelines that process large volumes of trade and financial data.

In my projects, data arrives from multiple upstream systems and is stored in Amazon S3. We use AWS Lambda for initial validation, AWS Glue Crawlers and the Glue Data Catalog for metadata management, and AWS Glue Spark jobs to cleanse, transform, deduplicate, and convert data into partitioned Parquet format. Business users query the curated data through Amazon Athena, while reporting datasets are loaded into Amazon Redshift. For larger Spark workloads requiring additional control, we use Amazon EMR.

I also monitor production environments using CloudWatch, troubleshoot ETL failures, optimise Spark performance through partitioning, broadcast joins, and Adaptive Query Execution, and implement incremental loading using Glue Bookmarks and watermark strategies. Security is maintained using IAM roles, KMS encryption, and least-privilege access.

What I enjoy most is solving production issues, improving pipeline reliability, and optimising performance while ensuring data accuracy. I'm looking forward to

bringing these skills to a role where I can build scalable, secure, and efficient cloud-based data engineering solutions."

Complete Interview Series (200 Questions)

You now have coverage across:

-  Amazon S3 (Basic → Advanced)
-  IAM & Security
-  AWS Lambda
-  AWS Glue (Basic → Advanced)
-  Glue Crawlers & Data Catalog
-  CloudWatch & CloudTrail
-  Athena (Basic → Advanced)
-  Amazon Redshift
-  Amazon EMR
-  Snowflake
-  Spark & PySpark
-  Partitioning & Performance Tuning
-  Incremental & Full Loads
-  CDC & SCD
-  Data Quality & Reconciliation
-  ETL Architecture
-  Production Support & Incident Management
-  Disaster Recovery & High Availability
-  Investment Banking Scenarios
-  Behavioural & Leadership Questions

This forms a comprehensive interview preparation guide for an experienced AWS Data Engineer, progressing from fundamentals to advanced production scenarios.

how do you load data into snowflake from s3 step by step explain in like separate paragraph

Below is a **real-time AWS Data Engineer interview answer** explaining **how to load data from Amazon S3 into Snowflake**. This is how you can answer in an interview for **2-5 minutes**.

Step 1: Source System Generates Data

The process starts with the source systems, such as trading applications, banking systems, ERP systems, or databases. These systems generate data in formats like CSV, JSON, Parquet, or Avro. Depending on the business requirement, files may be generated every few minutes, hourly, or as part of a daily batch process.

Step 2: Upload Files to Amazon S3

The generated files are uploaded into an Amazon S3 bucket, which acts as the landing or raw data layer. We organise the bucket using folders such as **raw**, **processed**, **archive**, and **error**. For example:

```
s3://bank-data/raw/trades/2026/07/29/trades_20260729.csv
```

In production, S3 is chosen because it provides scalable, durable, and cost-effective storage.

Step 3: Validate the Incoming File

Once the file arrives in S3, validation is performed before loading it into Snowflake. Validation checks include:

- File exists
- File size is greater than zero
- Correct file extension
- File naming convention
- Mandatory columns
- Duplicate file detection
- Schema validation
- Null value validation

If validation fails, the file is moved to an **Error** folder and an alert is sent to the support team.

Step 4: Trigger the Pipeline

After successful validation, the loading process is triggered. This can happen using:

- AWS Lambda
- AWS Glue Workflow
- Apache Airflow
- AWS Step Functions
- EventBridge
- Scheduled jobs

In many production environments, an S3 Event Notification triggers a Lambda function whenever a new file is uploaded.

Step 5: Create Snowflake Storage Integration

Instead of storing AWS access keys inside Snowflake, we create a **Storage Integration**. This allows Snowflake to securely access S3 using an IAM Role.

Example:

SQL

```
CREATE STORAGE INTEGRATION s3_integration
TYPE = EXTERNAL_STAGE
STORAGE_PROVIDER = S3
ENABLED = TRUE
STORAGE_AWS_ROLE_ARN = 'arn:aws:iam::123456789012:role/SnowflakeRole'
STORAGE_ALLOWED_LOCATIONS = ('s3://bank-data/raw/');
```

This is the recommended production approach because it avoids hardcoding AWS credentials.

Step 6: Configure IAM Role

The IAM Role grants Snowflake permission to read data from the S3 bucket. The role typically includes permissions such as:

- s3:GetObject
- s3:ListBucket

The bucket policy is also updated to trust the Snowflake IAM user or role. This enables secure communication between Snowflake and Amazon S3.

Step 7: Create an External Stage

An External Stage tells Snowflake where the files are stored.

Example:

SQL

```
CREATE STAGE trade_stage
URL='s3://bank-data/raw/trades/'
STORAGE_INTEGRATION = s3_integration;
```

The stage acts like a pointer to the S3 location and is reused whenever files need to be loaded.

Step 8: Create a File Format

Snowflake needs to know how to interpret the incoming files.

For CSV:

SQL

```
CREATE FILE FORMAT csv_format
TYPE = CSV
FIELD_OPTIONALLY_ENCLOSED_BY='"'
SKIP_HEADER=1;
```

For Parquet:

SQL

```
CREATE FILE FORMAT parquet_format
TYPE = PARQUET;
```

This ensures Snowflake reads the file correctly.

Step 9: Verify the Files

Before loading data, we verify that Snowflake can access the S3 files.

Example:

SQL

```
LIST @trade_stage;
```

This command lists all available files in the stage.

Step 10: Create the Target Table

The destination table must exist before loading.

Example:

SQL

```
CREATE TABLE trades
(
  trade_id INT,
  customer_id INT,
  trade_date DATE,
  amount NUMBER(12,2)
);
```

The table structure should match the incoming data.

Step 11: Load Data Using COPY INTO

The **COPY INTO** command loads data from S3 into Snowflake.

Example:

SQL

```
COPY INTO trades
FROM @trade_stage
FILE_FORMAT = (FORMAT_NAME='csv_format');
```

Snowflake automatically reads the files, parses the data, and inserts it into the target table.

Step 12: Validate the Load

After loading, we verify the data.

Typical validation includes:

SQL

```
SELECT COUNT(*) FROM trades;
```

We also compare:

- Source record count
- Target record count
- Null values
- Duplicate records
- Sample business data

This confirms that the load completed successfully.

Step 13: Handle Failed Records

If some records fail during loading, Snowflake provides load history and error information.

Useful commands include:

SQL

```
SELECT *  
FROM TABLE(INFORMATION_SCHEMA.LOAD_HISTORY());
```

and

SQL

```
VALIDATE(trades, JOB_ID => '_last');
```

These help identify rows rejected because of data type mismatches, invalid dates, or malformed records.

Step 14: Archive the Processed File

After a successful load, the processed file is moved from the **raw** folder to an **archive** folder.

Example:

```
raw/
```

↓

archive/

This prevents the same file from being loaded again and helps meet audit and retention requirements.

Step 15: Monitor the Pipeline

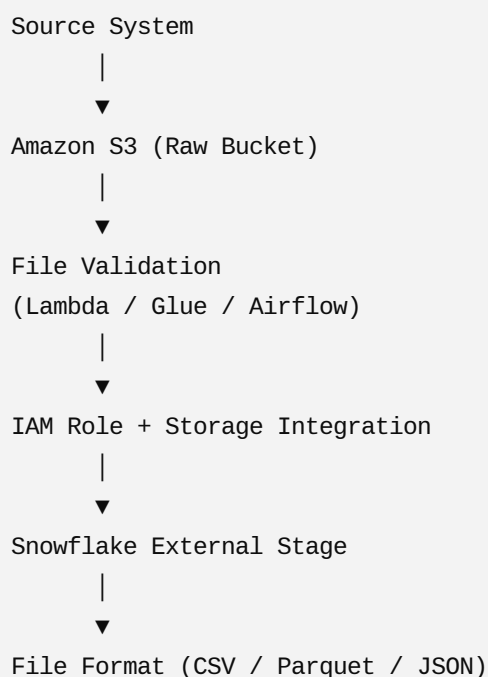
Finally, we monitor the end-to-end process.

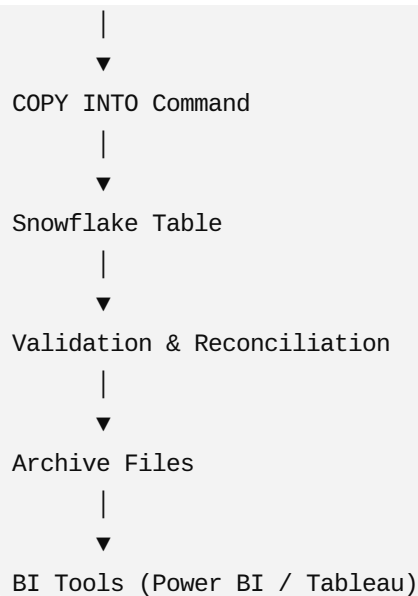
We monitor:

- File arrival in S3
- Lambda execution
- Glue or Airflow workflow status
- Snowflake COPY history
- Failed records
- Processing time
- Record counts
- Alerts and notifications

Monitoring ensures that any issues are detected quickly and resolved before they impact downstream reporting.

Complete End-to-End Architecture





Interview Summary (2-3 minutes)

"In our project, source systems generated trade files and stored them in Amazon S3. After the files arrived, we validated the filename, schema, mandatory columns, and duplicate status. We configured a Snowflake Storage Integration with an AWS IAM Role to provide secure access to S3 without embedding credentials. Next, we created an External Stage pointing to the S3 location and defined the appropriate file format. We then used the `COPY INTO` command to load the data into Snowflake tables. After loading, we validated record counts and business rules, checked the load history for any rejected records, archived the processed files to avoid duplicate processing, and continuously monitored the pipeline using AWS and Snowflake monitoring tools. This approach provided a secure, scalable, and reliable data loading process."

ChatGPT can make mistakes. Check important info.